Lab4-**2. Binary search: Find a target**

```
#include <stdio.h>
int binary_search(int values[], int n, int target) {
   int lo = 0;
   int hi = n;

   while (lo < hi) {
      int mid = lo + (hi - lo) / 2;

      if (values[mid] == target) {
         return mid;
      }

      if (values[mid] < target) {
         lo = mid + 1;
      } else {
         hi = mid;
      }
   }

   return -1;
}

// Test Binary Search
void test_search(const char *name, int values[], int n, int target, int expected) {
   int result = binary_search(values, n, target);
```

```
    printf("%-25s : ", name);

    if (result == expected) {
        printf("PASSED");
    } else {
        printf("FAILED");
    }

    printf(" | Target: %d | Result: %d | Expected: %d\n",
           target, result, expected);
}

int main() {
    int values[] = {-1, 0, 3, 5, 9, 12};
    int single[] = {7};
    int empty[1] = {0};

    printf("BINARY SEARCH TESTS:\n");



    test_search("Target found", values, 6, 9, 4);
    test_search("Target not found", values, 6, 2, -1);


    test_search("First index", values, 6, -1, 0);
    test_search("Last index", values, 6, 12, 5);
```

```
    test_search("One value - found", single, 1, 7, 0);
    test_search("One value - missing", single, 1, 5, -1);



    test_search("Below all values", values, 6, -5, -1);
    test_search("Above all values", values, 6, 15, -1);
    test_search("Between values", values, 6, 4, -1);



    test_search("Empty array", empty, 0, 5, -1);


    return 0;
}
```

```
BINARY SEARCH TESTS:
Target found              : PASSED | Target: 9 | Result: 4 | Expected: 4
Target not found          : PASSED | Target: 2 | Result: -1 | Expected: -1
First index               : PASSED | Target: -1 | Result: 0 | Expected: 0
Last index                : PASSED | Target: 12 | Result: 5 | Expected: 5
One value - found         : PASSED | Target: 7 | Result: 0 | Expected: 0
One value - missing       : PASSED | Target: 5 | Result: -1 | Expected: -1
Below all values          : PASSED | Target: -5 | Result: -1 | Expected: -1
Above all values          : PASSED | Target: 15 | Result: -1 | Expected: -1
Between values            : PASSED | Target: 4 | Result: -1 | Expected: -1
Empty array               : PASSED | Target: 5 | Result: -1 | Expected: -1
```